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SEMICONDUCTOR DEVICE AND SEMICONDUCTOR DEVICE MANUFACTURING METHOD

Abstract

A semiconductor device includes an insulating layer having a first surface and a second surface, a first semiconductor layer and a first thermally conductive portion provided in isolation from each other on the first surface side of the insulating layer, and a first element isolation layer defining the first semiconductor layer and covering the first thermally conductive portion. The first thermally conductive portion has a third surface facing the first semiconductor layer in a first direction parallel to the first surface and a fourth surface facing a second direction that is orthogonal to the first direction and being directed away from the first surface. In the semiconductor device, the thickness of a first part of the first element isolation layer located between the first semiconductor layer and the third surface is less than the thickness of the insulating layer between the first surface and the second surface.

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Background/Summary

BACKGROUND OF THE INVENTION

1. FIELD OF THE INVENTION

[0001] The embodiments discussed herein relate to a semiconductor device and a semiconductor device manufacturing method.

2. Description of the Prior Art

[0002] Semiconductor devices using substrates of silicon on insulator (SOI) structure are known.

[0003] For example, there is known a semiconductor device in which an island silicon region where a laterally diffused metal oxide semiconductor (LDMOS) transistor is formed is surrounded by an insulating film including a buried silicon oxide film and a silicon oxide film formed in a groove for element isolation so as to enhance breakdown voltage and element-isolating performance.

[0004] For such semiconductor devices, there are known a technique of making a part of a buried silicon oxide film thin, a technique of disposing polycrystalline silicon in the thin part, a technique of filling the interior of a silicon oxide film formed in a groove for element isolation with polycrystalline silicon, and others (see, for example, Japanese Laid-open Patent Publication No. H09-97832).

[0005] There is known a semiconductor device in which a substrate of SOI structure is used and a semiconductor layer in an active region where a transistor element is formed and an element isolation layer that is provided around the semiconductor layer to define the semiconductor layer are provided on the insulating layer of the substrate.

[0006] In such a semiconductor device, the thermal conductivities of materials used for the insulating layer and element isolation layer surrounding the semiconductor layer are lower than that of a material used for the semiconductor layer in the active region. Therefore, heat generated by self-heating in the semiconductor layer surrounded by the insulating layer and element isolation layer is difficult to dissipate to the outside. As a result, the temperature of the semiconductor layer increases, which may cause performance degradation of the semiconductor device.

[0007] In order to improve the heat dissipation, the thickness of the insulating layer provided under the semiconductor layer may be made thinner. In this case, however, there is a risk of lowering the breakdown voltage of the semiconductor device.

[0008] In addition, polycrystalline silicon or the like may be provided in the element isolation layer provided around the semiconductor layer. In this case, depending on its placement and connection with other portions, there are risks of reducing the layout flexibility for circuit elements such as electrodes and wiring and lowering the breakdown voltage.

SUMMARY OF THE INVENTION

[0009] According to one aspect, there is provided a semiconductor device, including: an insulating layer having a first surface on a first side thereof and a second surface on a second side thereof opposite to the first side; a first semiconductor layer provided on the first side of the insulating layer; a first thermally conductive portion provided in isolation from the first semiconductor layer on the first side of the insulating layer, the first thermally conductive portion having a third surface and a fourth surface, the third surface facing the first semiconductor layer in a first direction parallel to the first surface, the fourth surface facing a second direction that is orthogonal to the first

direction and is directed away from the first surface; and a first element isolation layer provided on the first side of the insulating layer to define the first semiconductor layer and to cover the third surface and the fourth surface of the first thermally conductive portion, wherein a thickness of a first part of the first element isolation layer located between the first semiconductor layer and the third surface is less than a thickness of the insulating layer between the first surface and the second surface.

[0010] The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

[0011] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention.

[0012] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a first view for describing an example of a semiconductor device according to a first embodiment;

[0014] FIG. 2 is a second view for describing the example of the semiconductor device according to the first embodiment;

[0015] FIG. 3 is a view for describing an example of a semiconductor device according to a second embodiment;

[0016] FIG. 4 is a view for describing an example of a semiconductor device according to a third embodiment;

[0017] FIG. 5 is a view for describing an example of a semiconductor device according to a fourth embodiment;

[0018] FIG. 6 is a first view for describing an example of a semiconductor device manufacturing method according to a fifth embodiment;

[0019] FIG. 7 is a second view for describing the example of the semiconductor device manufacturing method according to the fifth embodiment;

[0020] FIG. 8 is a third view for describing the example of the semiconductor device manufacturing method according to the fifth embodiment;

[0021] FIG. 9 is a fourth view for describing the example of the semiconductor device manufacturing method according to the fifth embodiment;

[0022] FIG. 10 is a fifth view for describing the example of the semiconductor device manufacturing method according to the fifth embodiment;

[0023] FIG. 11 is a sixth view for describing the example of the semiconductor device manufacturing method according to the fifth embodiment;

[0024] FIG. 12 is a seventh view for describing the example of the semiconductor device manufacturing method according to the fifth embodiment;

[0025] FIG. 13 is an eighth view for describing the example of the semiconductor device manufacturing method according to the fifth embodiment; and

[0026] FIG. 14 is a ninth view for describing the example of the semiconductor device manufacturing method according to the fifth embodiment.

DETAILED DESCRIPTION

First Embodiment

[0027] FIGS. 1 and 2 are views for describing an example of a semiconductor device according to a first embodiment. FIG. 1 is a schematic plan view of a main part of the example semiconductor

device. FIG. 2 is a schematic sectional view of the main part of the example semiconductor device, taken along a line II-II of FIG. 1.

[0028] In FIGS. 1 and 2, an example of heat **100** generated in the semiconductor device **1** and its propagation is schematically represented by thick arrows, for convenience.

[0029] The semiconductor device **1** illustrated in FIGS. 1 and 2 is an example of a semiconductor device having a transistor element formed using a substrate (referred to as an “SOI substrate”) of SOI structure. As an example, the illustrated semiconductor device **1** has a laterally diffused metal oxide semiconductor (LDMOS) transistor **10**, which is one type of high breakdown voltage transistor, as a transistor element.

[0030] As illustrated in FIGS. 1 and 2, the semiconductor device **1** includes an insulating layer **20**, a first semiconductor layer **30**, a first thermally conductive portion **40**, and a first element isolation layer **50**.

[0031] The insulating layer **20** has a first surface **20a** (upper surface) and a second surface **20b** (lower surface) opposite to the first surface **20a**, as illustrated in FIG. 2. A variety of insulating materials, for example, silicon oxide (SiO₂ or SiO), are used for the insulating layer **20**. The insulating layer **20** is a buried oxide (BOX) layer in the SOI substrate.

[0032] The first semiconductor layer **30** is provided on the first surface **20a** side of the insulating layer **20**, as illustrated in FIG. 2. For example, the first semiconductor layer **30** is provided in contact with the first surface **20a** of the insulating layer **20**, as illustrated in FIG. 2. A variety of semiconductor materials, for example, silicon (Si), are used for the first semiconductor layer **30**. For example, the first semiconductor layer **30** is formed by patterning a predetermined semiconductor material formed on the first surface **20a** side of the insulating layer **20** through etching.

[0033] The first element isolation layer **50** is provided on the first surface **20a** side of the insulating layer **20**, as illustrated in FIG. 2. The first element isolation layer **50** is provided to define the first semiconductor layer **30**, as illustrated in FIGS. 1 and 2. A variety of insulating materials, for example, silicon oxide, are used for the first element isolation layer **50**. For example, the first element isolation layer **50** is formed with shallow trench isolation (STI), which fills the area around the first semiconductor layer **30**, formed by patterning through etching, with a predetermined insulating material.

[0034] The LDMOS transistor **10** as illustrated in FIGS. 1 and 2 is formed on the first semiconductor layer **30** defined by the first element isolation layer **50**. The first semiconductor layer **30** is an active region or a device region where the LDMOS transistor **10** is formed in the semiconductor device **1**.

[0035] The LDMOS transistor **10** includes a body region **11** of a first conductivity type and a drift region **12** of a second conductivity type adjacent to the body region **11**, as illustrated in FIG. 2. The first conductivity type and the second conductivity type are opposite to each other. For example, the first conductivity type is p-type, and the second conductivity type is n-type.

[0036] A body contact region **13** of the first conductivity type and a source region **14** of the second conductivity type are provided at the surface layer of the body region **11**, as illustrated in FIGS. 1 and 2. A drain region **15** of the second conductivity type is provided at the surface layer of the drift region **12**, as illustrated in FIGS. 1 and 2. The source region **14** is isolated from the drift region **12** by the body region **11**. A deep well layer **16** of the second conductivity type is provided at the bottom layers of the body region **11** and drift region **12** located on the insulating layer **20** side, as illustrated in FIG. 2.

[0037] As illustrated in FIGS. 1 and 2, a gate electrode **18** is provided on the side of the body region **11** and drift region **12** opposite to the insulating layer **20**, that is, on the body region **11** and drift region **12** with a gate insulating film **17** interposed therebetween. A variety of insulating materials, for example, silicon oxide, are used for the gate insulating film **17**, and a variety of electrically conductive materials, for example, polysilicon, are used for the gate electrode **18**.

[0038] As illustrated in FIGS. 1 and 2, contacts **19** are connected to the body contact region **13**, source region **14**, drain region **15**, and gate electrode **18** (contacts connected to the gate electrode **18** are not illustrated). A variety of electrically conductive materials, for example, tungsten (W), are used for the contacts **19**. Although not illustrated here, the contacts **19** are electrically connected to a wiring layer of multilayer wiring structure that is provided to cover the LDMOS transistor **10**. [0039] In this connection, a metal silicide layer (not illustrated) may be provided on the surface layers of the body contact region **13**, source region **14**, drain region **15**, and gate electrode **18**, and the contacts **19** may be connected to the metal silicide layer.

[0040] A second element isolation layer **60** may be provided in the first semiconductor layer **30** in order to extend the drift distance of carriers in the drift region **12** between the gate electrode **18** and the drain region **15** and to thereby increase the breakdown voltage. The second element isolation layer **60** may be continuous with the first element isolation layer **50**. The second element isolation layer **60** may be formed as a part of the first element isolation layer **50**, that is, a part of the first element isolation layer **50** extending from just below the gate electrode **18** to reach the drain region **15**.

[0041] A second semiconductor layer **70** serving as a support substrate may be provided on the second surface **20b** side of the insulating layer **20**, as illustrated in FIG. 2. A variety of semiconductor materials, for example, silicon, are used for the second semiconductor layer **70**. For example, the second semiconductor layer **70** may be connected to the earth potential.

[0042] A laminate of the insulating layer **20** and first semiconductor layer **30** or a laminate of the insulating layer **20**, first semiconductor layer **30**, and second semiconductor layer **70** corresponds to the SOI substrate. In the SOI substrate, the insulating layer **20** is referred to as a BOX layer. The BOX layer is referred to as a buried insulating layer, an embedded insulating layer, or another.

[0043] For example, the LDMOS transistor **10** may be configured as an n-type LDMOS transistor, in which the above first conductivity type is p-type and the above second conductivity type is n-type. Alternatively, the LDMOS transistor **10** may be configured as a p-type LDMOS transistor, in which the above first conductivity type is n-type and the above second conductivity type is p-type.

[0044] The semiconductor device **1** is provided with the first thermally conductive portion **40**. The first thermally conductive portion **40** is provided on the first surface **20a** side of the insulating layer **20**, as illustrated in FIG. 2. The first thermally conductive portion **40** is provided in isolation from the first semiconductor layer **30**, as illustrated in FIGS. 1 and 2. As an example, the first thermally conductive portion **40** is provided to surround the entire periphery of the first semiconductor layer **30** where the LDMOS transistor **10** is formed, as illustrated in FIGS. 1 and 2.

[0045] For example, the first thermally conductive portion **40** is provided in contact with the first surface **20a** of the insulating layer **20** and in isolation from the first semiconductor layer **30**, as illustrated in FIG. 2. For example, the first thermally conductive portion **40** and the first semiconductor layer **30** are provided such that their surfaces facing toward the insulating layer **20** are in the same plane parallel to the first surface **20a** and are, in this example, flush with the first surface **20a**, as illustrated in FIG. 2.

[0046] A variety of materials with thermal conductivity are used for the first thermally conductive portion **40**. For example, a semiconductor material such as silicon is used for the first thermally conductive portion **40**. As an example, the same semiconductor material as used for the first semiconductor layer **30** is used for the first thermally conductive portion **40**. For example, silicon is used for both the first thermally conductive portion **40** and the first semiconductor layer **30**. For example, the first thermally conductive portion **40** is formed, together with the first semiconductor layer **30** in the device region, by patterning through etching.

[0047] As illustrated in FIGS. 1 and 2, the first thermally conductive portion **40** has a third surface **40a** (side surface) and a fourth surface **40b** (upper surface). The third surface **40a** faces the first semiconductor layer **30** in a first direction D1 parallel to the first surface **20a** of the insulating layer **20**, and the fourth surface **40b** faces a second direction D2 that is orthogonal to the first direction

D1 and is directed away from the first surface 20a.

[0048] As illustrated in FIGS. 1 and 2, the third surface 40a of the first thermally conductive portion 40 is provided at a position a predetermined distance apart from the first semiconductor layer 30. The predetermined distance here corresponds to the thickness T1 of a first part 51 of the first element isolation layer 50 that will be described later. As illustrated in FIG. 2, the height H1 of the fourth surface 40b of the first thermally conductive portion 40 from the first surface 20a of the insulating layer 20 is lower than the height H2, from the first surface 20a of the insulating layer 20, of a fifth surface 30a (upper surface) of the first semiconductor layer 30 located opposite to the insulating layer 20. In this connection, the fifth surface 30a of the first semiconductor layer 30 is not covered with the first element isolation layer 50 or second element isolation layer 60 but is exposed therefrom.

[0049] As illustrated in FIGS. 1 and 2, the first element isolation layer 50 is provided to cover the third surface 40a and fourth surface 40b of the first thermally conductive portion 40.

[0050] In this connection, the first thermally conductive portion 40 has, in addition to the third surface 40a, a surface 40c (side surface) opposite to the third surface 40a as surfaces facing the first direction D1 parallel to the first surface 20a of the insulating layer 20, as illustrated in FIGS. 1 and 2. The first element isolation layer 50 is provided to cover the surface 40c as well as the third surface 40a and fourth surface 40b of the first thermally conductive portion 40. It may be said that the first thermally conductive portion 40 is provided in the first element isolation layer 50.

[0051] In the semiconductor device 1, the thickness T1 of the first part 51 of the first element isolation layer 50 located between the first semiconductor layer 30 and the third surface 40a of the first thermally conductive portion 40 is less than the thickness T2 of the insulating layer 20 between the first surface 20a and the second surface 20b thereof, as illustrated in FIG. 2. In the semiconductor device 1, the distance between the first semiconductor layer 30 and the first thermally conductive portion 40 (the third surface 40a thereof) provided in isolation from the first semiconductor layer 30, i.e., the thickness T1 of the first part 51 of the first element isolation layer 50 located between the first semiconductor layer 30 and the first thermally conductive portion 40 is adjusted so as to obtain the thickness relationship of $T1 < T2$.

[0052] A second thermally conductive portion 80 that is continuous with the first thermally conductive portion 40 may be provided in the first element isolation layer 50, as illustrated in FIGS. 1 and 2. For example, the second thermally conductive portion 80 is provided to extend from the fourth surface 40b of the first thermally conductive portion 40 in the second direction D2 until the second thermally conductive portion 80 partly extends outside the first element isolation layer 50. As illustrated in FIGS. 1 and 2, a contact 19 formed of tungsten or the like may be connected to the part of the second thermally conductive portion 80 extending outside the first element isolation layer 50.

[0053] In the semiconductor device 1, the first thermally conductive portion 40, or the first thermally conductive portion 40 and second thermally conductive portion 80, when connected to the first thermally conductive portion 40, are provided so that they are electrically floating, i.e., they are electrically isolated from the LDMOS transistor 10 and other circuit elements of the semiconductor device 1.

[0054] In the semiconductor device 1 using the SOI substrate, the first semiconductor layer 30 in the device region where the LDMOS transistor 10 is formed is isolated from its surroundings by the insulating layer 20 and first element isolation layer 50. This secures the breakdown voltages between the LDMOS transistor 10 and the second semiconductor layer 70 provided on the second surface 20b side of the insulating layer 20 and between the LDMOS transistor 10 and another transistor or another that is provided with the first element isolation layer 50 interposed therebetween.

[0055] In the semiconductor device 1 configured as above, the LDMOS transistor 10 formed on the first semiconductor layer 30 in the device region generates heat during its operation. This heat

generation is referred to as self-heating.

[0056] In the above-described semiconductor device **1**, the thermal conductivities of the materials used for the insulating layer **20** and first element isolation layer **50** surrounding the first semiconductor layer **30** in the device region may be lower than that of the material used for the first semiconductor layer **30**. For example, in the case where silicon is used for the first semiconductor layer **30** and silicon oxide is used for the insulating layer **20** and first element isolation layer **50**, silicon has a thermal conductivity of 168 W/(m.Math.K), and silicon oxide has a thermal conductivity of 1.38 W/(m.Math.K).

[0057] The following describes a hypothetical case where the first thermally conductive portion **40** (and the second thermally conductive portion **80** connected thereto) as described above is not provided in the semiconductor device **1**. Even in this case, the first semiconductor layer **30** on which the LDMOS transistor **10** is formed is isolated from its surroundings by the insulating layer **20** and first element isolation layer **50**, which secures the breakdown voltage, as described above. However, the first semiconductor layer **30** is surrounded by the insulating layer **20** and first element isolation layer **50** that each has a lower thermal conductivity than the first semiconductor layer **30**, and therefore the heat generated during the operation of the LDMOS transistor **10** is difficult to be dissipated outside the first semiconductor layer **30**. If the heat dissipation is suppressed and the temperature of the first semiconductor layer **30** therefore increases, the driving capability of the LDMOS transistor **10** is reduced accordingly. If the thickness T2 of the insulating layer **20** is made thin, the heat dissipation from the first semiconductor layer **30** through the insulating layer **20** would increase. However, the reduction in the thickness T2 of the insulating layer **20** causes a decrease in the breakdown voltage. Especially, in the case where the second semiconductor layer **70** is provided on the side of the insulating layer **20** opposite to the first semiconductor layer **30** and the second semiconductor layer **70** is at the earth potential, the breakdown voltage is more likely to decrease. If the thickness T2 of the insulating layer **20** is made thick, such a decrease in the breakdown voltage would be avoided, but the heat dissipation from the first semiconductor layer **30** through the insulating layer **20** will decrease.

[0058] By contrast, in the above-described semiconductor device **1**, the first thermally conductive portion **40** is provided apart from the first semiconductor layer **30** in the device region where the LDMOS transistor **10** is formed, in the first element isolation layer **50**, as illustrated in FIGS. **1** and **2**. In addition, the second thermally conductive portion **80** may be further provided which is continuous with the first thermally conductive portion **40** in the first element isolation layer **50** and partly extends outside the first element isolation layer **50**. The first element isolation layer **50** is provided so as to cover the first thermally conductive portion **40**, that is, to cover the third surface **40a** and fourth surface **40b** (and surface **40c**) thereof. The first thermally conductive portion **40** and first element isolation layer **50** are provided such that the thickness T1 of the first part **51** of the first element isolation layer **50** located between the third surface **40a** of the first thermally conductive portion **40** and the first semiconductor layer **30** is less than the thickness T2 of the insulating layer **20**.

[0059] As a result, in the semiconductor device **1**, the breakdown voltage is secured by the relatively thick insulating layer **20** and first element isolation layer **50** that surround the first semiconductor layer **30**. In addition, in the semiconductor device **1**, the heat **100** generated by the self-heating of the LDMOS transistor **10** formed on the first semiconductor layer **30** is propagated through the relatively thin first part **51** to the first thermally conductive portion **40**, as illustrated in FIGS. **1** and **2**.

[0060] In the semiconductor device **1**, even while the relatively thick insulating layer **20** reduces the propagation and dissipation of the heat **100** from the first surface **20a** to the second surface **20b**, the heat **100** is propagated to the first thermally conductive portion **40** through the relatively thin first part **51**. The heat **100** propagated to the first thermally conductive portion **40** is propagated or diffused, or is dispersed inside the first thermally conductive portion **40**. This suppresses the

increase in the temperature of the first semiconductor layer **30** caused by the self-heating of the LDMOS transistor **10**, and accordingly suppresses the reduction in the driving capability of the LDMOS transistor **10** caused by the temperature increase.

[0061] In the semiconductor device **1**, the decrease in the breakdown voltage is suppressed by the relatively thick insulating layer **20**. Even in the case where the second semiconductor layer **70** is provided on the second surface **20b** side of the insulating layer **20** and is at the earth potential, the decrease in the breakdown voltage is suppressed. Furthermore, in the semiconductor device **1**, the first thermally conductive portion **40**, through which the heat **100** from the first semiconductor layer **30** is propagated, is electrically floating, which suppresses the decrease in the breakdown voltage effectively.

[0062] In the semiconductor device **1**, the first thermally conductive portion **40** and the second thermally conductive portion **80** that is continuous with the first thermally conductive portion **40** are further provided. Therefore, the heat **100** propagated from the first semiconductor layer **30** to the first thermally conductive portion **40** is further propagated or diffused, or is dispersed to the second thermally conductive portion **80**. This effectively suppresses the increase in the temperature of the first semiconductor layer **30** caused by the self-heating of the LDMOS transistor **10**. In addition, the second thermally conductive portion **80** has a part extending outside the first element isolation layer **50**, so that the heat **100** is dissipated from the part extending outside the first element isolation layer **50**. This further effectively suppresses the increase in the temperature of the first semiconductor layer **30** caused by the self-heating of the LDMOS transistor **10**. In addition, the second thermally conductive portion **80**, as well as the first thermally conductive portion **40**, is formed in an electrically floating state, which suppresses the decrease in the breakdown voltage.

[0063] In the semiconductor device **1**, a region where the second thermally conductive portion **80**, which is continuous with the first thermally conductive portion **40**, is provided may be called a heat sink region.

[0064] In the semiconductor device **1**, as illustrated in FIG. **1** (and FIG. **2**), the gate electrode **18** of the LDMOS transistor **10** overlaps the first semiconductor layer **30** in plan view. In the case where the second element isolation layer **60** is provided, the gate electrode **18** is provided to overlap the first semiconductor layer **30** and second element isolation layer **60** in plan view. Furthermore, the gate electrode **18** may be provided to overlap the first element isolation layer **50**, and the first thermally conductive portion **40** covered with the first element isolation layer **50** in plan view, as illustrated in FIG. **1**. In the semiconductor device **1**, the first element isolation layer **50** is provided to cover the first thermally conductive portion **40**. This avoids a reduction in the layout flexibility for the gate electrode **18** in the case where the gate electrode **18** is provided (or routed) on the first element isolation layer **50**. In other words, restrictions on the layout (or routing) of the gate electrode **18** due to the presence of the first thermally conductive portion **40** are eased.

[0065] The case of providing the gate electrode **18** has been described as an example. Alternatively, the semiconductor device **1** may be provided with a variety of circuit elements, for example, an electrode or a part thereof, a wiring or a part thereof, a resistor or a part thereof, a capacitor or a part thereof, and an inductor or a part thereof on the first element isolation layer **50**. Such cases suppress the reduction in the layout flexibility for the variety of circuit elements as well. That is, the semiconductor device **1** may have a layout in which such a circuit element is provided to overlap the first element isolation layer **50** and the first thermally conductive portion **40** covered with the first element isolation layer **50** in plan view.

[0066] With the above-described configuration, the semiconductor device **1** is achieved, which has high layout flexibility for circuit elements and suppresses performance degradation caused by the self-heating and a decrease in the breakdown voltage.

[0067] In addition, with regard to the semiconductor device **1** configured as described above, the first thermally conductive portion **40** is provided in contact with the first surface **20a** of the insulating layer **20**, for example, as illustrated in FIG. **2**. In the case where depressions are formed

in the first surface **20a** of the insulating layer **20**, the first thermally conductive portion **40** may be provided in contact with the inner surfaces of the depressions in the first surface **20a**.

[0068] Furthermore, with respect to the semiconductor device **1**, the first semiconductor layer **30** is provided in contact with the first surface **20a** of the insulating layer **20**, for example, as illustrated in FIG. **2**. In the case where depressions are provided in the first surface **20a** of the insulating layer **20**, the first semiconductor layer **30** may be provided in contact with the inner surfaces of the depressions in the first surface **20a**.

[0069] In addition, with respect to the semiconductor device **1**, the first thermally conductive portion **40** and the first semiconductor layer **30** do not necessarily have to be in contact with the first surface **20a** of the insulating layer **20**, and another layer (an insulating layer or another) may be interposed between them. The first thermally conductive portion **40** and the first semiconductor layer **30** may be provided such that their surfaces facing toward the insulating layer **20** are located in the same plane parallel to the first surface **20a**, regardless of whether or not another layer is interposed between each of the first thermally conductive portion **40** and first semiconductor layer **30** and the first surface **20a** of the insulating layer **20**.

[0070] Further, with respect to the semiconductor device **1**, the first element isolation layer **50** may be provided to have a second part **52** extending further toward the first semiconductor layer **30** than the first part **51**, for example, as illustrated in FIGS. **1** and **2**. The second part **52** is a part that is provided on the side of the first part **51** opposite to the insulating layer **20** and is continuous with the first part **51**. The second part **52** has a sixth surface **50a** (lower surface) that faces toward the insulating layer **20** and is located apart from the first surface **20a** of the insulating layer **20**, as illustrated in FIG. **2**. A part of the first semiconductor layer **30** is interposed between the sixth surface **50a** and the first surface **20a** and is in contact with the sixth surface **50a**. The first element isolation layer **50** may be provided such that the height of the sixth surface **50a** from the first surface **20a**, that is, the height, from the first surface **20a**, of the part of the first semiconductor layer **30** interposed between the sixth surface **50a** and the first surface **20a** is equal to the height **H1** of the fourth surface **40b** of the first thermally conductive portion **40** from the first surface **20a**.

[0071] Still further, with respect to the semiconductor device **1**, in the case where the second element isolation layer **60** as illustrated in FIGS. **1** and **2** is provided in the first semiconductor layer **30**, the second element isolation layer **60** is provided to have a seventh surface **60a** (lower surface) that faces toward the insulating layer **20** and is located apart from the first surface **20a** of the insulating layer **20**, as illustrated in FIG. **2**. A part of the first semiconductor layer **30** is interposed between the seventh surface **60a** and the first surface **20a** and is in contact with the seventh surface **60a**. The second element isolation layer **60** may be provided such that the height of the seventh surface **60a** from the first surface **20a**, that is, the height, from the first surface **20a**, of the part of the first semiconductor layer **30** interposed between the seventh surface **60a** and the first surface **20a** is equal to the height **H1** of the fourth surface **40b** of the first thermally conductive portion **40** from the first surface **20a**.

[0072] Still further, with respect to the semiconductor device **1**, the first surface **20a** and second surface **20b** of the insulating layer **20** do not necessarily have to be flat surfaces, but may be surfaces with concave portions, convex portions, concave and convex portions, or the like.

[0073] Still further, with respect to the semiconductor device **1**, the third surface **40a** of the first thermally conductive portion **40** does not necessarily have to be a flat surface perpendicular to the first surface **20a** of the insulating layer **20**, but may be a surface with concave portions, convex portions, concave and convex portions, or the like. Even in the case where the third surface **40a** is not a flat surface perpendicular to the first surface **20a**, the heat **100** is able to be propagated through the first part **51** of the first element isolation layer **50** located between the first semiconductor layer **30** and the third surface **40a**, which makes it possible to suppress an increase in the temperature of the first semiconductor layer **30**.

[0074] Still further, with respect to the semiconductor device **1**, the fourth surface **40b** of the first

thermally conductive portion **40** does not necessarily have to be a flat surface parallel to the first surface **20a** of the insulating layer **20**, but may be a surface with concave portions, convex portions, concave and convex portions, or the like. Even in the case where the fourth surface **40b** is not a flat surface parallel to the first surface **20a**, it is possible to cover the first thermally conductive portion **40** with the first element isolation layer **50** and to dispose a variety of circuit elements on the first element isolation layer **50** covering the first thermally conductive portion **40**.

[0075] Still further, with respect to the semiconductor device **1**, the fifth surface **30a** of the first semiconductor layer **30** does not necessarily have to be a flat surface parallel to the first surface **20a** of the insulating layer **20**, but may be a surface with concave portions, convex portions, concave and convex portions, or the like.

[0076] Still further, with respect to the semiconductor device **1**, the sixth surface **50a** of the second part **52** of the first element isolation layer **50** (or a partial surface of the first semiconductor layer **30** in contact therewith) does not necessarily have to be a flat surface parallel to the first surface **20a** of the insulating layer **20**, but may be a surface with concave portions, convex portions, concave and convex portions, or the like.

[0077] Still further, with respect to the semiconductor device **1**, the seventh surface **60a** of the second element isolation layer **60** (or a partial surface of the first semiconductor layer **30** in contact therewith) does not necessarily have to be a flat surface parallel to the first surface **20a** of the insulating layer **20**, but may be a surface with concave portions, convex portions, concave and convex portions, or the like.

[0078] In FIG. **2**, the insulating layer **20** is illustrated thinner than the first semiconductor layer **30** for convenience. However, the thickness relationship between the first semiconductor layer **30** and the insulating layer **20** is not limited to the illustrated one.

[0079] Although not illustrated here, the semiconductor device **1** may be further provided with a region where an alignment mark, which is used in a process of manufacturing the semiconductor device **1**, is formed. The details of the process of manufacturing the semiconductor device **1** will be described in detail later (fifth embodiment). In the semiconductor device **1**, the region where such an alignment mark is formed may be called an alignment mark region.

Second Embodiment

[0080] The first embodiment has described, as an example, the semiconductor device **1** that includes the LDMOS transistor **10**, which is one type of high breakdown voltage transistor, as a transistor element. However, the type of the transistor element is not limited thereto.

[0081] FIG. **3** illustrates an example of a semiconductor device according to a second embodiment. FIG. **3** is a schematic sectional view of a main part of the example semiconductor device.

[0082] In FIG. **3**, an example of heat **100A** generated in the semiconductor device **1A** and its propagation is schematically represented by thick arrows, for convenience.

[0083] The semiconductor device **1A** illustrated in FIG. **3** differs from the semiconductor device **1** described above in the first embodiment in that the semiconductor device **1A** is configured to have a MOS transistor **10A** on a first semiconductor layer **30** in the device region defined by a first element isolation layer **50**. The remaining configuration of the semiconductor device **1A** is the same as that of the above-described semiconductor device **1**.

[0084] The MOS transistor **10A** includes a source region **14A**, a drain region **15A**, a gate insulating film **17A**, a gate electrode **18A**, and a sidewall insulating film **17Aa**. The source region **14A** and the drain region **15A** are provided in isolation from each other in the first semiconductor layer **30**. The gate electrode **18A** is provided on the first semiconductor layer **30** between the source region **14A** and the drain region **15A**, with the gate insulating film **17A** interposed therebetween, and the sidewall insulating film **17Aa** is provided on the sidewall of the gate electrode **18A**. Contacts **19** are connected to the source region **14A**, drain region **15A**, and gate electrode **18A** (contacts connected to the gate electrode **18A** are not illustrated).

[0085] As in this semiconductor device **1A**, a transistor element configured like the MOS transistor

10A, which is not limited to high voltage applications, may be formed in the device region.

[0086] The MOS transistor **10A** generates heat during its operation. In the semiconductor device **1A**, the heat **100A** generated by the MOS transistor **10A** formed on the first semiconductor layer **30** is propagated to a first thermally conductive portion **40** through a first part **51** of the first element isolation layer **50**, which has a thickness **T1** less than the thickness **T2** of the insulating layer **20**. The heat **100A** propagated to the first thermally conductive portion **40** is further propagated to a second thermally conductive portion **80** that is continuous with the first thermally conductive portion **40**, and is dissipated through a part of the second thermally conductive portion **80** extending outside the first element isolation layer **50** or a contact **19** connected to the extending part. As a result, an increase in the temperature of the first semiconductor layer **30** due to the self-heating of the MOS transistor **10A** is suppressed, and a reduction in the driving capability of the MOS transistor **10A** caused by the temperature increase is suppressed accordingly.

[0087] In addition, the semiconductor device **1A** is also configured so that the first semiconductor layer **30**, on which the MOS transistor **10A** is formed, is surrounded by the insulating layer **20** and the first element isolation layer **50**, which suppresses a decrease in the breakdown voltage. Furthermore, the semiconductor device **1A** is also configured so that the first thermally conductive portion **40** is covered with the first element isolation layer **50**, which suppresses a reduction in the layout flexibility for a variety of circuit elements to be provided on the first element isolation layer **50**, including the gate electrode **18A**.

[0088] This second embodiment also achieves the semiconductor device **1A** that has high layout flexibility for circuit elements and suppresses performance degradation caused by the self-heating and a decrease in the breakdown voltage.

[0089] In FIG. **3**, the insulating layer **20** is illustrated thinner than the first semiconductor layer **30** for convenience. However, the thickness relationship between the first semiconductor layer **30** and the insulating layer **20** is not limited to the illustrated one.

Third Embodiment

[0090] The first embodiment has described, as an example, the semiconductor device **1** that includes the first thermally conductive portion **40** surrounding the entire periphery of the first semiconductor layer **30** in plan view. However, the location and shape of the first thermally conductive portion **40** is not limited thereto.

[0091] FIG. **4** illustrates an example of a semiconductor device according to a third embodiment. FIG. **4** is a schematic plan view of a main part of the example semiconductor device.

[0092] The semiconductor device **1B** illustrated in FIG. **4** differs from the semiconductor device **1** described above in the first embodiment in that the semiconductor device **1B** is configured so that a first thermally conductive portion **40** is provided in a first element isolation layer **50** to surround only a part of the entire periphery of a first semiconductor layer **30** on which an LDMOS transistor **10** is formed. The remaining configuration of the semiconductor device **1B** is the same as that of the above-described semiconductor device **1**.

[0093] As in this semiconductor device **1B**, the first thermally conductive portion **40** does not necessarily have to surround the entire periphery of the first semiconductor layer **30**.

[0094] In the semiconductor device **1B** as well, the heat generated by the LDMOS transistor **10** is propagated to the first thermally conductive portion **40** through a relatively thin first part **51** of the first element isolation layer **50**. The heat propagated to the first thermally conductive portion **40** is further propagated to a second thermally conductive portion **80** that is continuous with the first thermally conductive portion **40**, and is dissipated through a part of the second thermally conductive portion **80** extending outside the first element isolation layer **50** or a contact **19** connected to the extending part. As a result, an increase in the temperature of the first semiconductor layer **30** due to the self-heating of the LDMOS transistor **10** is suppressed, and a reduction in the driving capability of the LDMOS transistor **10** caused by the temperature increase is suppressed accordingly.

[0095] Furthermore, as to the layout flexibility for a variety of circuit elements and the breakdown voltage, the semiconductor device **1B** has the same effects as the above-described semiconductor device **1**.

[0096] That is, this third embodiment also achieves the semiconductor device **1B** that has high layout flexibility for circuit elements and suppresses performance degradation caused by the self-heating and a decrease in the breakdown voltage.

Fourth Embodiment

[0097] The first embodiment has described, as an example, the semiconductor device **1** that has one LDMOS transistor **10** as a transistor element. However, the semiconductor device **1** may be configured so that a plurality of transistor elements are integrated.

[0098] FIG. **5** is a view for describing an example of a semiconductor device according to a fourth embodiment. FIG. **5** is a schematic plan view of a main part of the example semiconductor device.

[0099] The semiconductor device **1C** illustrated in FIG. **5** includes a first semiconductor layer **30** that is provided on an insulating layer **20** of SOI substrate in the device region defined by a first element isolation layer **50**, as in the case described above. As an example, two LDMOS transistors **10** that are connected in parallel and share a drift region **12** (not illustrated) and a drain region **15**, two sets of complementary metal oxide semiconductor (CMOS) transistors **200**, and two wirings **260** are provided in the device region.

[0100] Each CMOS transistor **200** includes an n-channel MOS (nMOS) transistor **200n** and a p-channel MOS (pMOS) transistor **200p**.

[0101] The nMOS transistor **200n** has an n-type source region **210n**, an n-type drain region **220n** adjacent to the n-type source region **210n**, and a p-type body region (p-type well region) **230p** facing the n-type source region **210n** and n-type drain region **220n**. The nMOS transistor **200n** also has a separate gate **240** located between the n-type source region **210n** and n-type drain region **220n** on one side and the p-type body region **230p** on the other, and a gate electrode **250** located between the n-type source region **210n** and the n-type drain region **220n**. Contacts **19** are connected to the n-type source region **210n**, n-type drain region **220n**, and p-type body region **230p**.

[0102] The pMOS transistor **200p** has a p-type source region **210p**, a p-type drain region **220p** adjacent to the p-type source region **210p**, and an n-type body region (n-type well region) **230n** facing the p-type source region **210p** and p-type drain region **220p**. The pMOS transistor **200p** also has a separate gate **240** located between the p-type source region **210p** and p-type drain region **220p** on one side and the n-type body region **230n** on the other, and a gate electrode **250** located between the p-type source region **210p** and the p-type drain region **220p**. Contacts **19** are connected to the p-type source region **210p**, the p-type drain region **220p**, and the n-type body region **230n**.

[0103] Each transistor element of the LDMOS transistor **10**, nMOS transistor **200n**, and pMOS transistor **200p**, which are configured as above, is formed in the corresponding device region defined by the first element isolation layer **50**.

[0104] In the semiconductor device **1C**, a first thermally conductive portion **40** is provided to surround the two LDMOS transistors **10** in plan view. In the semiconductor device **1C**, first thermally conductive portions **40** are provided to surround the nMOS transistors **200n** and pMOS transistors **200p** of the CMOS transistors **200**, respectively, in plan view. In addition, the first thermally conductive portions **40** surrounding the transistor elements of the LDMOS transistor **10**, nMOS transistor **200n**, and pMOS transistor **200p** are connected by an additional first thermally conductive portion **40**. The first element isolation layer **50** is provided to cover the first thermally conductive portions **40** provided as above. Second thermally conductive portions **80** are connected at predetermined positions, in this example, two positions to the first thermally conductive portions **40**, and a contact **19** is connected to a part of each second thermally conductive portion **80** extending outside the first element isolation layer **50**.

[0105] The gate electrodes **18** of the LDMOS transistors **10**, as well as wirings **18a** drawn from the gate electrodes **18** and contacts **19** connected thereto, are provided on the first element isolation

layer **50** covering the first thermally conductive portions **40**. In addition, the separate gates **240** and gate electrodes **250** of the CMOS transistors **200**, as well as wirings **250a** drawn from the gate electrodes **250** and contacts **19** connected thereto, are provided on the first element isolation layer **50**. Furthermore, a conductor pattern of a predetermined shape, in this example, two straight wirings **260** and contacts **19** connected thereto are provided on the first element isolation layer **50**. For example, the wirings **260**, which form a conductor pattern, may be formed of the same material as the gate electrodes **18**, the wirings **18a** drawn from the gate electrodes **18**, the separate gates **240** and gate electrodes **250**, and the wirings **250a** drawn from the gate electrodes **250**. The wirings **260** may be used not only as mere wirings but also as, for example, resistors. For example, the gate electrodes **18**, the wirings **18a** drawn from the gate electrodes **18**, the separate gates **240**, the gate electrodes **250**, the wirings **250a** drawn from the gate electrodes **250**, and the wirings **260** may be formed of polycrystalline silicon. By adjusting the widths and lengths of the wirings **260** appropriately, the wirings **260** may be used as resistive elements. The wirings **260**, when formed of polycrystalline silicon, may be used as polycrystalline silicon resistors.

[0106] During the operation of the semiconductor device **1C**, each transistor element of the LDMOS transistors **10**, nMOS transistors **200n**, and pMOS transistors **200p** generates heat. The heat generated by each transistor element is propagated through a relatively thin first part **51** of the first element isolation layer **50** to the first thermally conductive portions **40** and further to the second thermally conductive portions **80**, and is then dissipated through the contacts **19** connected thereto. As a result, an increase in the temperature of the first semiconductor layer **30** due to the self-heating of each transistor element is suppressed, and a reduction in the driving capability of each transistor element caused by the temperature increase is suppressed accordingly.

[0107] In addition, in the semiconductor device **1C**, the first thermally conductive portions **40**, which surround each transistor element of the LDMOS transistors **10**, nMOS transistors **200n**, and pMOS transistors **200p** in plan view and are connected with each other, are covered with the first element isolation layer **50**. Therefore, in the semiconductor device **1C**, a reduction in the layout flexibility for a variety of circuit elements that are provided on the first element isolation layer **50** is suppressed. In this example, the variety of circuit elements includes the gate electrodes **18**, wirings **18a**, separate gates **240**, gate electrodes **250**, wirings **250a**, and wirings **260**.

[0108] In this example, the gate electrodes **18**, wirings **18a**, separate gates **240**, gate electrodes **250**, wirings **250a**, and wirings **260** are provided as circuit elements that are provided on the first element isolation layer **50**. In addition to these, a variety of circuit elements may be provided on the first element isolation layer **50**, such as a capacitor having a dielectric between a pair of electrodes or a part thereof, and an inductor with a conductor formed in a predetermined pattern shape or a part thereof, as well as a resistor formed using polycrystalline silicon or the like described above as an example or a part thereof. Even in the case where such circuit elements are provided, the reduction in the layout flexibility is suppressed, as in the case described above.

[0109] In the semiconductor device **1C** as well, the first semiconductor layer **30** on which the transistor elements of the LDMOS transistors **10**, nMOS transistors **200n**, and pMOS transistors **200p** are formed is surrounded by the insulating layer **20** and the first element isolation layer **50**, which suppresses a decrease in the breakdown voltage.

[0110] This fourth embodiment also achieves the semiconductor device **1C** that have high layout flexibility for circuit elements and suppresses performance degradation caused by the self-heating and a decrease in the breakdown voltage.

Fifth Embodiment

[0111] The following describes an example of a method of manufacturing a semiconductor device as a fifth embodiment. Here, a method of manufacturing the semiconductor device **1** having the LDMOS transistor **10** as described above in the first embodiment will be described as an example.

[0112] FIGS. **6** to **14** are views for describing an example of a semiconductor device manufacturing method according to the fifth embodiment. FIGS. **6** to **14** are schematic sectional views of a main

part in each process for manufacturing the semiconductor device. The following describes the processes for manufacturing the semiconductor device one by one, with reference to FIGS. 6 to 14. [0113] FIG. 6 is a view for describing an example of a first etching process.

[0114] First, an SOI substrate is prepared. FIG. 6 illustrates an SOI substrate in which the first semiconductor layer **30** (semiconductor material) is provided on the first surface **20a** side of the insulating layer **20** (BOX layer) and the second semiconductor layer **70** (support substrate or semiconductor substrate) is provided on the second surface **20b** side of the insulating layer **20**. The insulating layer **20** is, for example, a silicon oxide layer with the thickness **T2** of 300 nm (0.3 μm) to 600 nm (0.6 μm). The first semiconductor layer **30** is, for example, a silicon layer with a thickness **T3** of 100 nm to 150 nm. The second semiconductor layer **70** is, for example, a silicon substrate.

[0115] In FIG. 6, the insulating layer **20** is illustrated thinner than the first semiconductor layer **30** for convenience. However, the thickness relationship between the first semiconductor layer **30** and the insulating layer **20** is not limited to the illustrated one. The same holds true for the thickness relationship between the first semiconductor layer **30** and the insulating layer **20** in FIGS. 7 to 14 that will be described below.

[0116] On the front surface of the first semiconductor layer **30** of the SOI substrate, a silicon oxide film (not illustrated) with a thickness of 7 nm to 14 nm is formed by hydrochloric acid oxidation at a temperature of 900° C. to 1000° C. Then, a resist **400** is formed on the silicon oxide film as illustrated in FIG. 6. The resist **400** is formed by exposure using a mask having openings in predetermined regions and development after the exposure. Using the resist **400** as a mask, the silicon oxide film is anisotropically etched with a mixed gas of trifluoromethane (CHF₃) and tetrafluoromethane (CF₄). In addition, using the resist **400** and the etched silicon oxide film as masks, the first semiconductor layer **30** is anisotropically etched to a depth of about 50 nm to 70 nm with a chlorine (Cl₂)-based gas. As a result, a first trench **401** for defining the device region **300** and the heat sink region **301**, and a second trench **402** serving as a mark for an alignment mark region **302** are formed, as illustrated in FIG. 6.

[0117] After the first trench **401** and the second trench **402** are formed, the resist **400** is removed by ashing. The silicon oxide film formed on the front surface of the first semiconductor layer **30** remains.

[0118] FIG. 7 is a view for describing an example of a deep well layer forming process.

[0119] A resist **403** is formed on the first semiconductor layer **30** (silicon oxide film remaining on the front surface thereof) etched as described above, as illustrated in FIG. 7. The resist **403** has an opening in a region where the deep well layer **16** is to be formed. The resist **403** is formed by photolithography using the second trench **402** formed in the alignment mark region **302** for alignment.

[0120] Using the formed resist **403** as a mask, an n-type impurity, for example, phosphorus (P) is ion-implanted to form an n-type deep well layer **16** at the bottom layer of the first semiconductor layer **30**, as illustrated in FIG. 7. As an example, phosphorus is ion-implanted under the conditions of an acceleration energy of 1.5 MeV to 2.5 MeV, inclusive, and an implantation amount (dose) of 4.5×10^{12} cm^{sup.2} to 5.5×10^{12} cm^{sup.2}, inclusive, to thereby form the n-type deep well layer **16**.

[0121] After the deep well layer **16** is formed, the resist **403** is removed by ashing.

[0122] FIG. 8 is a view for describing an example of a second etching process.

[0123] After the deep well layer **16** is formed, a silicon oxide film (not illustrated) with a thickness of 7 nm to 14 nm is formed on the front surface of the first semiconductor layer **30** by hydrochloric acid oxidation at a temperature of 900° C. to 1000° C. Then, a silicon nitride film **404** with a thickness of 100 nm to 150 nm is formed on the silicon oxide film by the chemical vapor deposition (CVD) method, as illustrated in FIG. 8. After that, a resist (not illustrated) is formed on the silicon nitride film **404**.

[0124] After that, the formed resist is exposed using a mask having an opening for forming an element isolation layer pattern and is then developed. Then, the silicon nitride film **404** and the silicon oxide film are anisotropically etched using the developed resist as a mask, with a mixed gas of trifluoromethane and tetrafluoromethane.

[0125] Then, the resist is removed by ashing. The first semiconductor layer **30** is selectively anisotropically etched using the silicon nitride film **404** remaining after the removal of the resist as a mask, with a chlorine-based gas until the insulating layer **20** is exposed. By doing so, a trench **405** that defines the first semiconductor layer **30** in the device region **300** is formed, and the first thermally conductive portion **40**, which is provided in isolation from the first semiconductor layer **30**, is formed in the trench **405**, as illustrated in FIG. 8. In addition, the second thermally conductive portion **80** that is continuous with the first thermally conductive portion **40** is formed in the heat sink region **301**. For example, the first thermally conductive portion **40** is formed to surround the entire periphery of the first semiconductor layer **30** in the device region **300** defined by the trench **405** in plan view, and the second thermally conductive portion **80** is formed to be continuous with a part of the first thermally conductive portion **40**. In addition, an alignment mark **406** is formed in the alignment mark region **302**.

[0126] The trench **405** includes a trench for forming the first element isolation layer **50**, which will be described later, and a trench for forming the second element isolation layer **60**, which will be described later.

[0127] The first thermally conductive portion **40** has the third surface **40a**, which serves as a side surface facing the first semiconductor layer **30** in the device region **300**, and the fourth surface **40b**, which serves as an upper surface. The trench **405** includes a first isolating trench portion **405a** the first semiconductor layer **30** in the device region **300** from the third surface **40a** of the first thermally conductive portion **40**. The trench **405** also includes a second trench portion **405b** that is provided on the side of the first trench portion **405a** opposite to the insulating layer **20** and that is continuous with the first trench portion **405a** and extends further toward the first semiconductor layer **30** in the device region **300** than the first trench portion **405a**.

[0128] The width between the first semiconductor layer **30** in the device region **300** and the third surface **40a** of the first thermally conductive portion **40**, that is, the width **W1** of the first trench portion **405a** (corresponding to the thickness **T1** of the first part **51** of the first element isolation layer **50** to be described later) is set to a value ranging from 0.15 μm to 0.2 μm . The width **W1** of the first trench portion **405a** is set to a value smaller than the thickness **T2** of the insulating layer **20** ranging from 0.3 μm (300 nm) to 0.6 μm (600 nm). The width of the first trench **401** described above in the process of FIG. 6 is adjusted so that the first trench portion **405a** has such a width **W1** through the etching.

[0129] The fourth surface **40b** of the first thermally conductive portion **40** is located at the bottom of the second trench portion **405b**. The height **H1** of the bottom of the second trench portion **405b** from the first surface **20a**, i.e., the height **H1** of the fourth surface **40b** of the first thermally conductive portion **40** from the first surface **20a** is lower than the height **H2**, from the first surface **20a**, of the fifth surface **30a** of the first semiconductor layer **30** located opposite to the insulating layer **20**. The second trench portion **405b** and the first thermally conductive portion **40** are formed by etching so as to obtain such a height **H1**.

[0130] In this connection, the bottom of the second trench portion **405b** of the trench **405** serves as the sixth surface **50a** of the second part **52** of the first element isolation layer **50**, which will be described later. The bottom of the trench **405** that is not continuous with the first trench portion **405a** serves as the seventh surface **60a** of the second element isolation layer **60**, which will be described later.

[0131] The alignment mark **406** has an eighth surface **406a** that serves as its upper surface. The eighth surface **406a** is located at the bottom of the second trench portion **405b**. For example, the height of the eighth surface **406a** from the first surface **20a** is equal to the height **H1** of the bottom

of the second trench portion **405b** from the first surface **20a**, that is, the height **H1** of the fourth surface **40b** of the first thermally conductive portion **40** from the first surface **20a**. For example, the second trench portion **405b**, the first thermally conductive portion **40**, and the alignment mark **406** are formed by etching so as to obtain such a height **H1**.

[0132] In this connection, not limited to the above example, the trench **405** does not necessarily have to be formed such that the heights of the fourth surface **40b** of the first thermally conductive portion **40** and the eighth surface **406a** of the alignment mark **406** with respect to the first surface **20a** of the insulating layer **20** match each other.

[0133] FIG. **9** is a view for describing an example of an element isolation layer forming process.

[0134] A silicon oxide film (not illustrated) with a thickness of 7 nm to 14 nm is formed on the front surface of the first semiconductor layer **30** etched as described above, by thermal oxidation at a temperature of 1050° C. to 1150° C. Then, a silicon oxide film with a thickness of 250 nm to 400 nm is further formed on the silicon oxide film by the CVD method to fill the trench **405**. The silicon oxide film filling the trench **405** is polished until the silicon nitride film **404** is exposed. As a result, the first element isolation layer **50** and the second element isolation layer **60** as illustrated in FIG. **9** are formed.

[0135] The formed first element isolation layer **50** fills the first and second trench portions **405a** and **405b** of the trench **405** and covers the third surface **40a** and fourth surface **40b** of the first thermally conductive portion **40**. The first part **51** of the first element isolation layer **50** is formed in the first trench portion **405a**, and the second part **52** of the first element isolation layer **50** is formed in the second trench portion **405b**. The thickness **T1** (corresponding to the width **W1** of the above-described first trench portion **405a**) of the first part **51** of the first element isolation layer **50** is less than the thickness **T2** of the insulating layer **20**. The bottom of the second part **52** of the first element isolation layer **50** serves as the sixth surface **50a** of the first element isolation layer **50**. The bottom of the second element isolation layer **60** serves as the seventh surface **60a** of the second element isolation layer **60**. A part of the first semiconductor layer **30** is interposed between each of the sixth and seventh surfaces **50a** and **60a** and the first surface **20a** and is in contact with the sixth surface **50a** and the seventh surface **60a**.

[0136] For example, the heights of the sixth surface **50a** of the first element isolation layer **50** and the seventh surface **60a** of the second element isolation layer **60** from the first surface **20a** of the insulating layer **20** are equal to the height **H1** of the fourth surface **40b** of the first thermally conductive portion **40** from the first surface **20a**. For example, in the above-described process of FIG. **8**, the trench **405** is formed by etching so as to form the first element isolation layer **50** having such a sixth surface **50a** and the second element isolation layer **60** having such a seventh surface **60a**.

[0137] After the first element isolation layer **50** and the second element isolation layer **60** are formed, the silicon nitride film **404** is removed with a boiling phosphoric acid solution.

[0138] In this connection, not limited to the above example, the first thermally conductive portion **40**, the first element isolation layer **50**, and the second element isolation layer **60** do not necessarily have to satisfy the condition that the heights of the fourth surface **40b**, sixth surface **50a**, and seventh surface **60a** with respect to the first surface **20a** of the insulating layer **20** match each other. In other words, not limited to the above example, the trench **405** does not necessarily have to be formed such that the heights of the fourth surface **40b**, sixth surface **50a**, and seventh surface **60a** with respect to the first surface **20a** of the insulating layer **20** match each other.

[0139] FIG. **10** is a view for describing an example of a body region forming process.

[0140] After the first element isolation layer **50** and the second element isolation layer **60** are formed and the silicon nitride film **404** is removed, a resist **407** having an opening in the region where the body region **11** of the first semiconductor layer **30** is to be formed is formed, as illustrated in FIG. **10**.

[0141] A p-type impurity, for example, boron (B) is ion-implanted using the formed resist **407** as a

mask, to form the p-type body region **11** in the first semiconductor layer **30**, as illustrated in FIG. **10**. As an example, boron is ion-implanted under the conditions of an acceleration energy of 100 keV to 200 keV, inclusive, and an implantation amount of 1.0×10^{12} cm.^{sup.}-2 to 3.0×10^{13} cm.^{sup.}-2, inclusive, to thereby form the p-type body region **11**.

[0142] A p-type impurity may be further ion-implanted into the body region **11** for controlling the threshold voltage of the transistor. As an example, for the threshold voltage control, boron is ion-implanted with the conditions of an acceleration energy of 5 keV to 30 keV, inclusive, and an implantation amount of 9.5×10^{12} cm.^{sup.}-2 to 1.05×10^{13} cm.^{sup.}-2, inclusive.

[0143] After the body region **11** is formed, the resist **407** is removed by ashing.

[0144] FIG. **11** is a view for describing an example of a drift region forming process.

[0145] After the body region **11** is formed, a resist **408** having an opening in the region where the drift region **12** of the first semiconductor layer **30** is to be formed is formed, as illustrated in FIG. **11**.

[0146] An n-type impurity, for example, phosphorus, arsenic (As), or the like is ion-implanted using the resist **408** as a mask, to form the n-type drift region **12** in the first semiconductor layer **30**, as illustrated in FIG. **11**. As an example, phosphorus is ion-implanted under the conditions of an acceleration energy of 0.5 MeV to 1.0 MeV, inclusive, and an implantation amount of 4.5×10^{12} cm.^{sup.}-2 to 5.5×10^{12} cm.^{sup.}-2, inclusive, to thereby form the n-type drift region **12**.

[0147] After the drift region **12** is formed, the resist **408** is removed by ashing.

[0148] FIG. **12** is a view for describing an example of a gate electrode forming process.

[0149] After the drift region **12** is formed, the silicon oxide film is removed by wet etching using a hydrofluoric acid. After that, the gate insulating film **17** is formed on the exposed surface of the first semiconductor layer **30** (body region **11** and drift region **12**) in the device region **300** by a thermal oxidation method, as illustrated in FIG. **12**. The thickness of the gate insulating film **17** is between 12 nm and 25 nm, inclusive, for example.

[0150] After the gate insulating film **17** is formed, a polysilicon film is deposited by the CVD method. The thickness of the polysilicon film is between 100 nm and 200 nm, inclusive, for example. After the polysilicon film is formed, a resist **409** is formed on the polysilicon film by photolithography, and the polysilicon film is patterned by dry etching using the resist **409** as a mask. As a result, the gate electrode **18** is formed.

[0151] The gate electrode **18** is formed to overlap the first semiconductor layer **30** and the second element isolation layer **60** in plan view. In addition, the gate electrode **18** is formed to overlap the first thermally conductive portion **40** in plan view, for example, as illustrated in FIG. **1**.

[0152] After the gate electrode **18** is formed, the resist **409** is removed by ashing.

[0153] In this connection, in addition to the gate electrode **18**, a wiring that is connected thereto or separated therefrom may be formed on the first element isolation layer **50**. Alternatively, a resistor or a part thereof, a capacitor or a part thereof, and/or an inductor or a part thereof may be formed on the first element isolation layer **50**. A variety of circuit elements such as the gate electrode **18** or another gate electrode, a wiring, a resistor, a capacitor, and an inductor may be formed to overlap the first thermally conductive portion **40** in plan view.

[0154] FIG. **13** is a view for describing an example of an impurity region and contact forming process.

[0155] After the gate electrode **18** is formed, the sidewall insulating film **17a** is formed on the sidewall of the gate electrode **18** by, for example, a silicon oxide film or the like, as illustrated in FIG. **13**.

[0156] Then, an n-type impurity, for example, phosphorus, arsenic, or the like is ion-implanted into the first semiconductor layer **30** in the device region **300** to form the n-type source region **14** in the surface layer of the body region **11** and the drain region **15** in the surface layer of the drift region **12**, as illustrated in FIG. **13**. As an example, phosphorus is ion-implanted under the conditions of

an acceleration energy of 12 keV to 18 keV, inclusive, and an implantation amount of 2.0×10^{15} cm.^{sup.}-2 to 6.0×10^{15} cm.^{sup.}-2, inclusive, to thereby form the n-type source region **14** and drain region **15**.

[0157] In addition, a p-type impurity, for example, boron or the like is further ion-implanted into the surface layer of the body region **11** to form the p-type body contact region **13**, as illustrated in FIG. **13**. As an example, boron is ion-implanted under the conditions of an acceleration energy of 4 keV to 10 keV, inclusive, and an implantation amount of 1.0×10^{15} cm.^{sup.}-2 to 4.0×10^{15} cm.^{sup.}-2, inclusive, to thereby form the p-type body contact region **13**.

[0158] After the ion implantation, heat treatment is performed in an inert gas atmosphere to activate the impurities implanted into the body contact region **13**, source region **14**, and drain region **15**.

[0159] For example, a metal silicide layer (not illustrated) is formed on the surface layers of the body contact region **13**, source region **14**, and drain region **15**.

[0160] After the body contact region **13**, the source region **14**, and the drain region **15** are formed, an interlayer dielectric film **410** is deposited as illustrated in FIG. **13**. Then, openings are formed in the interlayer dielectric film **410** to be continuous at predetermined locations with the body contact region **13**, source region **14**, drain region **15**, gate electrode **18** (not illustrated), and second thermally conductive portion **80**, and the openings are filled with an electrically conductive material such as tungsten to form the contacts **19** as illustrated in FIG. **13**.

[0161] FIG. **14** is a view for describing an example of a wiring layer forming process.

[0162] After the body contact region **13**, source region **14** drain region **15**, and then the interlayer dielectric film **410** and contacts **19** are formed, the interlayer dielectric film **410** is further deposited, as illustrated in FIG. **14**. Then, a wiring layer **411** is formed using an electrically conductive material such as copper (Cu) in the interlayer dielectric film **410**, for example, by the damascene method. The wiring layer **411** is connected to the contacts **19** formed in the device region **300** and the heat sink region **301**.

[0163] Through the processes described above, the semiconductor device **1** with the n-type LDMOS transistor **10** is manufactured.

[0164] The above description has described the method of manufacturing the semiconductor device **1** with the n-type LDMOS transistor **10**, as an example. Alternatively, a semiconductor device **1** with a p-type LDMOS transistor **10** may be similarly manufactured by swapping the p-type impurity and the n-type impurity in the above example.

[0165] As described earlier, in the semiconductor device **1**, the heat generated by the LDMOS transistor **10** in the device region **300** is easily propagated to the first thermally conductive portion **40** through the first part **51** of the first element isolation layer **50**, which is relatively thin compared to the insulating layer **20**. The heat propagated to the first thermally conductive portion **40** is dissipated through the second thermally conductive portion **80** continuous with the first thermally conductive portion **40** in the heat sink region **301**, and also through the contacts **19** and the wiring layer **411** connected thereto. This suppresses a reduction in the driving capability caused by the self-heating of the LDMOS transistor **10**.

[0166] In addition, in the semiconductor device **1**, the LDMOS transistor **10** is surrounded by the insulating layer **20** and the first element isolation layer **50**, and the insulating layer **20** may be made relatively thick, which suppresses a decrease in the breakdown voltage.

[0167] Furthermore, the first thermally conductive portion **40** is covered with the first element isolation layer **50**, which suppresses a reduction in the layout flexibility for a variety of circuit elements to be provided on the first element isolation layer **50**. With the above-described manufacturing method, the semiconductor device **1** is achieved, which has high layout flexibility for circuit elements and suppresses performance degradation caused by the self-heating and a decrease in the breakdown voltage.

[0168] The method of manufacturing the semiconductor device **1** described above in the first embodiment has been described as an example. Alternatively, it is also possible to employ the

example as described in the fifth embodiment to manufacture the semiconductor device **1A** (FIG. **3**) that has the MOS transistor **10A** instead of the LDMOS transistor **10** as described above in the second embodiment. It is also possible to employ the example as described in the fifth embodiment to manufacture the semiconductor device **1B** (FIG. **4**) that has the first thermally conductive portion **40** that does not surround the entire periphery of the first semiconductor layer **30** in plan view as described above in the third embodiment. It is also possible to employ the example as described in the fifth embodiment to manufacture the semiconductor device **1C** in which the LDMOS transistors **10** and the CMOS transistors **200** are integrated as described above in the fourth embodiment.

[0169] In one aspect, it is possible to achieve a semiconductor device that has high layout flexibility for circuit elements and suppresses performance degradation caused by self-heating and a decrease in breakdown voltage.

[0170] All examples and conditional language provided herein are intended for the pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although one or more embodiments of the present invention have been described in detail, it should be understood that various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

[0171] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor device, comprising: an insulating layer having a first surface on a first side thereof and a second surface on a second side thereof opposite to the first side; a first semiconductor layer provided on the first side of the insulating layer; a first thermally conductive portion provided in isolation from the first semiconductor layer on the first side of the insulating layer, the first thermally conductive portion having a third surface and a fourth surface, the third surface facing the first semiconductor layer in a first direction parallel to the first surface, the fourth surface facing a second direction that is orthogonal to the first direction and is directed away from the first surface; and a first element isolation layer provided on the first side of the insulating layer to define the first semiconductor layer and to cover the third surface and the fourth surface of the first thermally conductive portion; wherein a thickness of a first part of the first element isolation layer located between the first semiconductor layer and the third surface is less than a thickness of the insulating layer between the first surface and the second surface.
2. The semiconductor device according to claim 1, wherein the first thermally conductive portion surrounds an entire periphery of the first semiconductor layer in a plan view of the semiconductor device.
3. The semiconductor device according to claim 1, wherein a height of the fourth surface from the first surface is lower than a height, from the first surface, of a fifth surface of the first semiconductor layer located opposite to the insulating layer.
4. The semiconductor device according to claim 1, further comprising a second thermally conductive portion provided in the first element isolation layer, the second thermally conductive portion being continuous with the first thermally conductive portion and having a part extending outside the first element isolation layer.
5. The semiconductor device according to claim 1, further comprising a second semiconductor

layer provided on the second side of the insulating layer.

6. The semiconductor device according to claim 5, wherein the second semiconductor layer is connected to an earth potential.

7. The semiconductor device according to claim 1, wherein the first thermally conductive portion is electrically floating.

8. The semiconductor device according to claim 1, wherein the first thermally conductive portion is in contact with the first surface.

9. The semiconductor device according to claim 1, wherein the first thermally conductive portion and the first semiconductor layer have respective surfaces facing toward the insulating layer, and the respective surfaces are in a same plane parallel to the first surface.

10. The semiconductor device according to claim 1, wherein a same semiconductor material is used for both the first semiconductor layer and the first thermally conductive portion.

11. The semiconductor device according to claim 1, wherein: the first element isolation layer includes a second part extending further toward the first semiconductor layer than the first part, the second part having a sixth surface that faces toward the insulating layer and is located apart from the first surface; and a part of the first semiconductor layer is interposed between the sixth surface and the first surface and is in contact with the sixth surface.

12. The semiconductor device according to claim 11, wherein a height of the sixth surface from the first surface is equal to a height of the fourth surface from the first surface.

13. The semiconductor device according to claim 1, further comprising a second element isolation layer provided in the first semiconductor layer, the second element isolation layer having a seventh surface that faces toward the insulating layer and is located apart from the first surface, wherein a part of the first semiconductor layer is interposed between the seventh surface and the first surface and is in contact with the seventh surface.

14. The semiconductor device according to claim 13, wherein a height of the seventh surface from the first surface is equal to a height of the fourth surface from the first surface.

15. The semiconductor device according to claim 13, further comprising a gate electrode provided on a side of the first semiconductor layer opposite to the insulating layer so as to overlap the first semiconductor layer and the second element isolation layer in a plan view of the semiconductor device.

16. The semiconductor device according to claim 1, further comprising a circuit element provided on a side of the first element isolation layer opposite to the insulating layer so as to overlap the first thermally conductive portion in a plan view of the semiconductor device.

17. The semiconductor device according to claim 16, wherein the circuit element is at least one of an electrode, a wiring, a resistor, a capacitor, or an inductor.

18. The semiconductor device according to claim 1, further comprising an alignment mark provided on the first side of the insulating layer, the alignment mark having an eighth surface, wherein a height of the eighth surface from the first surface is equal to a height of the fourth surface from the first surface.

19. A semiconductor device manufacturing method, comprising: forming a first semiconductor layer on a first side of an insulating layer having a first surface on the first side thereof and a second surface on a second side thereof opposite to the first side; forming a first thermally conductive portion in isolation from the first semiconductor layer on the first side of the insulating layer, the first thermally conductive portion having a third surface and a fourth surface, the third surface facing the first semiconductor layer in a first direction parallel to the first surface, the fourth surface facing a second direction that is orthogonal to the first direction and is directed away from the first surface; and forming a first element isolation layer on the first side of the insulating layer to define the first semiconductor layer and to cover the third surface and the fourth surface of the first thermally conductive portion; wherein a thickness of a first part of the first element isolation layer located between the first semiconductor layer and the third surface is less than a thickness of

the insulating layer between the first surface and the second surface.

20. The semiconductor device manufacturing method according to claim 19, wherein the forming of the first thermally conductive portion includes forming the first thermally conductive portion so as to surround an entire periphery of the first semiconductor layer in a plan view.

21. The semiconductor device manufacturing method according to claim 19, wherein: the forming of the first thermally conductive portion includes forming a trench to define the first semiconductor layer by etching a semiconductor material provided on the first side of the insulating layer, and forming, in the trench, the first thermally conductive portion in isolation from the first semiconductor layer, and the forming of the first element isolation layer includes forming, in the trench, the first element isolation layer to cover the first thermally conductive portion.

22. The semiconductor device manufacturing method according to claim 21, wherein the forming of the first thermally conductive portion in the trench includes forming the first thermally conductive portion such that a height of the fourth surface from the first surface is lower than a height, from the first surface, of a fifth surface of the first semiconductor layer located opposite to the insulating layer.

23. The semiconductor device manufacturing method according to claim 21, wherein: the forming of the trench includes forming a first trench portion isolating the first semiconductor layer from the third surface, and forming a second trench portion on a side of the first trench portion opposite to the insulating layer such that the second trench portion is continuous with the first trench portion and extends further toward the first semiconductor layer than the first trench portion, and the forming of the first element isolation layer includes: forming the first part in the first trench portion, and forming a second part in the second trench portion such that the second part has a sixth surface facing toward the insulating layer and being located apart from the first surface, and a part of the first semiconductor layer is interposed between the sixth surface and the first surface and is in contact with the sixth surface.

24. The semiconductor device manufacturing method according to claim 19, further comprising forming a second thermally conductive portion in the first element isolation layer such that the second thermally conductive portion is continuous with the first thermally conductive portion and has a part extending outside the first element isolation layer.

25. The semiconductor device manufacturing method according to claim 19, further comprising forming a second element isolation layer in the first semiconductor layer such that the second element isolation layer has a seventh surface facing toward the insulating layer and being located apart from the first surface, and a part of the first semiconductor layer is interposed between the seventh surface and the first surface and is in contact with the seventh surface.

26. The semiconductor device manufacturing method according to claim 25, further comprising forming a gate electrode on a side of the first semiconductor layer opposite to the insulating layer such that the gate electrode overlaps the first semiconductor layer and the second element isolation layer in a plan view.

27. The semiconductor device manufacturing method according to claim 19, further comprising forming a circuit element on a side of the first element isolation layer opposite to the insulating layer such that the circuit element overlaps the first thermally conductive portion in a plan view.

28. The semiconductor device manufacturing method according to claim 27, wherein the circuit element is at least one of an electrode, a wiring, a resistor, a capacitor, or an inductor.

29. The semiconductor device manufacturing method according to claim 19, further comprising forming an alignment mark on the first side of the insulating layer such that the alignment mark has an eighth surface and a height of the eighth surface from the first surface is equal to a height of the fourth surface from the first surface.
